Working Title

How to use this template

Throughout this template, **grey text** provides instructions and guiding questions for each section. **Blue text** shows concrete examples of what a strong answer looks like — use it as a reference, do not copy it. Phrases highlighted in yellow are key phrases you can use in your proposal. Italic grey placeholders such as *<your text here>* mark where your own writing should go.

Submission

Write your proposal by filling in the placeholder text throughout the document. When you are ready to submit, change `\instructionstrue` to `\instructionsfalse` at the top of the `.tex` file (line 13), recompile, and submit the resulting **PDF**. The proposal must not exceed **3 pages** excluding references.

References

References are **not** counted toward the 2-page limit. Add your references to the `references.bib` file. Each entry has a unique key (e.g. `devlin2019bert`) that you use in `\citep{}` and `\citet{}` commands. The reference list below is generated automatically from that file when you compile.

How to cite in the text:

- Parenthetical: `\citep{key}` produces (Author, Year)
- Author as subject: `\citet{key}` produces Author (Year)
- Multiple references: `\citep{key1,key2}` produces (Author1, Year1; Author2, Year2)

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1. Basics

Project topic: *<Project topic assigned from the TAR 2026 topic list.>*

Team: *<Team name.>*

Team members: *<Full Name>; <JMBAG>*
<Full Name>; <JMBAG>
<Full Name>; <JMBAG>

2. Elevator Pitch

Describe your project in 2–3 sentences as if you had 30 seconds to explain it to someone unfamiliar with the topic. Name the **problem**, the **approach**, and the expected **contribution** — in that order. Write this section **last**, even though it appears first. It is the abstract of your abstract.

(1) What problem does your work address, and why does it matter?

"Despite the rapid growth of NLP benchmarks, most evaluate models in English only, leaving multilingual generalization largely unmeasured."

(2) What is your approach?

"We investigate whether cross-lingual transfer from pre-trained multilingual models can close this gap on low-resource Slavic languages."

(3) What will your work contribute?

"Our study produces a new benchmark and a set of empirical findings that will inform future multilingual model design."

With the advent of social media, the emergence of conspiracy theories has become frequent. We investigate the phenomenon of conspiracy in the context of Large Language Models. Our study will produce new empirical findings that help advance this field.

3. Motivation

If you can bring references to support your claims, do that. Feel free to use the phrases highlighted in yellow as is or their paraphrases.

Start big. What is the broad, general problem you are addressing? The bigger the better.

"The number of NLP publications has grown exponentially over the past decade (Joshi et al., 2020; Conneau et al., 2020), and language diversity remains a fundamental challenge for the field."

What particular, focused problem are you addressing?

"Despite this growth, the main goal addressed by this paper is understanding to what extent current pre-trained models generalize across typologically different languages without task-specific fine-tuning data."

Why is this an important problem?

"It is an important goal because the majority of the world's languages are low-resource, and excluding them from NLP research perpetuates a language divide with real societal consequences (Joshi et al., 2020; Lauscher et al., 2020)."

Why is this a non-trivial, hard problem?

"This is hard to do because language typology, script systems, and data availability interact in complex ways that make it difficult to predict when and why transfer succeeds or fails."

Misinformation is a big problem in today's world. One specific type of misinformation that stands out is conspiracy theories. A conspiracy theory is a piece of text that contains very radical conclusions which aren't built on a rational basis. An interesting problem regarding conspiracy theories in NLP is: "Can conspiracy theories be characterized only by their semantic structure?" At first glance , it appears not, but there exists research such as (Fort et al., 2023) that makes this a promising path to take. This is also a relevant problem since conspiracy theories evolve according to the current state of the world. It would be very beneficial to produce a model that can classify a conspiracy theory only based on its semantic structure. Since we plan on using modern methods of LLMs, a question of pretraining data also arises. Will our model work better if we additionally pretrain it on a conspiracy theory corpus?

4. Gap

Show what was done and what is missing. Think wider than just your topic: what is the broad class of problems your work is related to? Usually you will identify 3–4 relevant papers, which you will use as a starting point for your project. **This step is critical.** Do a thorough and systematic related work analysis *before* starting work on the project, to avoid solving a non-existing problem. The gap must be clear; continue monitoring related work as you proceed. If things change, revise your plan.

What has been done in the past?

"Pre-trained multilingual models such as mBERT and XLM-R have shown strong cross-lingual transfer on high-resource language pairs (Devlin et al., 2019; Conneau et al., 2020; Hu et al., 2020)."

What is the problem with existing work?

"However, evaluations are dominated by a small set of languages (Joshi et al., 2020). Low-resource Slavic languages are rarely included (Hu et al., 2020; Lauscher et al., 2020). Existing benchmarks do not control for domain shift, making it hard to isolate the contribution of cross-lingual transfer from domain mismatch (Conneau et al., 2020). Thus, no single existing study provides a comprehensive picture of transfer in this setting."

What is the missing part?

"Paper X evaluates cross-lingual transfer but only on high-resource languages. Paper Y covers low-resource languages but lacks domain control. Paper Z does both, but the data is not made public. To the best of our knowledge, our work is the first to evaluate cross-lingual transfer systematically across seven low-resource Slavic languages under controlled domain conditions."

Analysis of the semantic structure of conspiracy theories has been a common problem in psychoanalysis. (Fort et al., 2023) addresses this problem by training the model on data in which topical words have been masked. The issue with this paper is the usage of outdated models such as BART and SVM. There is potential to build on their work by using LLMs. (Platt et al., 2022) is based on a similar idea. The goal of that paper is to differentiate between conspiracy theory text and more general misinformation text. Another interesting research direction deals with pretraining . (Levy et al., 2021) investigates GPT- 2's ability to generate conspiracy text. They explain this is a consequence of conspiracy text being deeply rooted in the pretraining data. The problem with this paper is that it deals with the outdated GPT-2.

5. Research Questions & Contributions

List your research questions (RQs). An RQ is a question your study will answer — what will it add to our knowledge compared to related work? Good RQs start with: "What is the relationship between…", "What are the factors that affect…", "What types of X exist…" (exploratory), "How to do X…" (engineering). Avoid "Can X do Y?" unless you are confident it can, or will provide strong theoretical or empirical evidence that it cannot.

For each RQ, list your planned **contributions** — the tangible, reusable results your study will produce. A contribution can be a framework, a dataset, a model, a thorough analysis, or a research agenda. You will typically have 1–3 contributions.

"To enable the study of cross-lingual transfer in low-resource Slavic languages, this paper addresses three research questions and makes the following contributions:

RQ1: How well does zero-shot cross-lingual transfer from XLM-R generalize to low-resource Slavic languages on NER?
Contribution: We report a systematic evaluation across seven languages and three domains, providing the most comprehensive picture to date of transfer performance in this setting. More efficient.

RQ2: What linguistic factors predict transfer success or failure?
Contribution: We introduce a diagnostic framework that correlates model errors with morphological complexity, script type, and training data size.

RQ3: How to improve transfer for the lowest-resource languages?
Contribution: We propose and evaluate a lightweight adaptation strategy using automatically translated pivot data. Our results will pave the path for more inclusive multilingual NLP benchmarking."

RQ1:

Is it possible to detect a conspiracy theory if given a text in which actors, actions, etc. are masked?

Contribution:

This will give us a better insight in conspiracy theory texts. We will determine if conspiracy theries follow specific forms that are independent of factual truth and that can be used to differentiate between critically written text and a conspiracy theory.

RQ2:

Should the pretraining set contain conspiracy theories?

Contribution:

We aim to determine if additional pretraining on harmful data such as conspiracy theories can be done in a safe way which also improves the model. If proven possible this could lead to other research regarding AI safety

RQ3:

Can we addionally fine-tune the model to be resistant to sarcasm?

Contribution:

<What tangible result will this produce?>

RQ4:

How does learning conspiracy theories affect models performance on other tasks?

Contribution:

<What tangible result will this produce?>

6. Experimental Setup

Briefly describe your experimental plan across the four areas below. Each choice must be **justified** with respect to your RQs. For example, saying "we will use BERT" without explaining why teaches nothing. This section should be concrete enough to show the pipeline is feasible, but you do not need a full methods section. Close with a short paragraph on risks.

Datasets. What data will you use? Is it publicly available? Describe its key characteristics (size, domain, language, annotation scheme). If you plan to collect or synthesize data, briefly explain how.

"We will use the WikiAnn dataset (Pan et al., 2017), which provides NER annotations for 176 languages including all seven target Slavic languages. It is publicly available, large-scale, and has been used as a standard benchmark in prior cross-lingual transfer work (Hu et al., 2020; Lauscher et al., 2020), making our results directly comparable."

LOCO (description of LOCO)

Models & Baselines. What systems will you build, fine-tune, or prompt? What baselines will you compare against and why? Justify why these models are appropriate for your RQs.

"Our primary model is XLM-R_{large} (Conneau et al., 2020), chosen for its strong multilingual coverage of Slavic scripts. We compare against mBERT (Devlin et al., 2019) as a weaker multilingual baseline and a language-specific monolingual model as an upper-bound oracle. This setup directly addresses RQ1 by isolating the effect of multilingual pre-training scale."

some model that is easy to additionaly pretrain. (Fort et al., 2023) paper is our baseline

Evaluation. What metrics will you use to measure performance? Why are these metrics appropriate for your RQs? Note any known limitations of these metrics.

"We report span-level F_1 following the CoNLL evaluation standard, which is the de facto metric for NER and enables direct comparison with prior work. We also report per-language and per-entity-type breakdowns to address RQ2. We note that F_1 does not distinguish between boundary and label errors, which we will supplement with a qualitative error analysis."

For sure the same metrics as (Fort et al., 2023) (so we can compare results), but there should be some discussion regarding evaluation on other tasks

General Methodology. Describe your experimental pipeline in 3–5 steps. Walk the reader through how you will go from raw data to an answer for each RQ. This does not need to be exhaustive — just enough to show the plan is concrete and well thought through.

"(1) Preprocess and reformat WikiAnn splits for all seven target languages. (2) Fine-tune XLM-R and mBERT on the English training split only (zero-shot transfer setup). (3) Evaluate all models on each target language test set and record F_1 per language and entity type (RQ1). (4) Correlate error rates with language-level features such as morphological complexity and script type (RQ2). (5) Apply the pivot-data adaptation strategy and re-evaluate on the three lowest-resource languages (RQ3)."

You can write something about masking

Risks. What could go wrong? For each risk, state the impact on your RQs, and your mitigation or fallback plan.

"*Data availability:* WikiAnn coverage for two target languages is sparse (<1k sentences); if quality is insufficient we will fall back to silver data generated via projection from a related high-resource language. *Compute:* Fine-tuning XLM-R_{large} requires significant GPU memory; if unavailable we will use the base variant, which has been shown to close 85% of the gap at a fraction of the cost."

I am sure you can think of something.

7. Justification for API Credits

Write your justification for using the API credits as a mini-pitch — be **specific**. "We need the API to generate synthetic training data for low-resource language X because no open dataset exists" is compelling. "We want to try a large model" is not. You must also include a **fallback plan** — your project must be viable regardless of whether credits are awarded.

Use case: what specifically would you use the API for, and which RQ does it serve?

"We plan to use the API to generate silver-standard NER annotations for three languages for which no labeled data exists, directly enabling RQ3."

Alternative plan: if you do not receive credits, what is your fallback?

"We will restrict the study to the four languages for which labeled data already exists, narrowing the scope of RQ3 but keeping the core contributions intact."

<Your justification, following the structure above. Approx. 150 words.>

References

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